

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 10: Isosceles Triangles

Every formula and result of the chapter on one sheet

1	Theorem 1	Equal sides have equal opposite angles
2	Theorem 2	Equal angles have equal opposite sides
3	Base angle from vertical angle v	$\frac{180^\circ - v}{2}$
4	Vertical angle from base angle b	$180^\circ - 2b$
5	Altitude from the vertex	Bisects the base and the vertical angle
6	Equilateral triangle	Each angle 60°
7	Isosceles right triangle	Angles $90^\circ, 45^\circ, 45^\circ$
8	Height of an equilateral triangle	$\frac{\sqrt{3}}{2} \times \text{side}$
9	Altitude of an isosceles triangle	$\sqrt{a^2 - \left(\frac{b}{2}\right)^2}$, equal sides a , base b
10	Exterior angle	Sum of the two interior opposite angles
11	Equal segments	Altitudes, medians and angle bisectors to the equal sides are equal